

Exp no:09

Date :

Using Process Explorer to Identify Suspicious Processes

Aim

To use Microsoft Sysinternals Process Explorer to monitor system activities and identify any suspicious or malicious processes running on a Windows computer.

Requirements

- Windows operating system
- Internet connection
- Process Explorer (from Microsoft Sysinternals)
- Optional: Antivirus software (e.g., Windows Defender, Malwarebytes)

Description

- Process Explorer is a part of the **Microsoft Sysinternals Suite**. It is a powerful tool used to view detailed information about system processes. It helps investigators and administrators analyze active processes, detect suspicious behavior, monitor CPU and memory usage, and verify process authenticity using digital signatures.

Step-by-Step Process

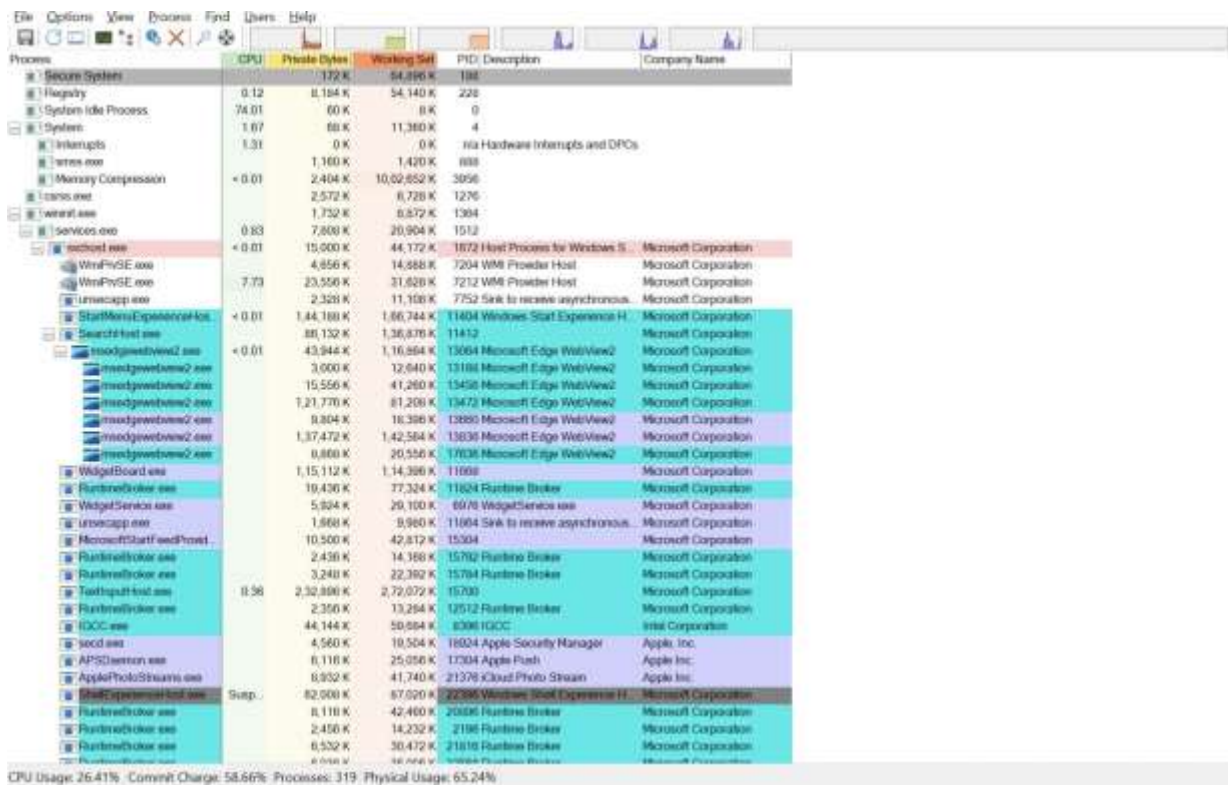
Step 1: Download and Setup Process Explorer

1. Go to the official Microsoft Sysinternals website:
 <https://learn.microsoft.com/en-us/sysinternals/downloads/process-explorer>
 2. Click **Download Process Explorer**.
 3. Extract the downloaded ZIP file to a folder.
 4. Right-click procexp64.exe (for 64-bit) or procexp.exe (for 32-bit) → select **Run as Administrator**.
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Step 2: Understand the Interface

1. The main window displays all running processes in a **hierarchical tree view**.
2. Each process shows details such as **PID, CPU usage, memory usage, and company name**.
3. Color codes represent process states:

-  **Green** — Newly started processes
-  **Red** — Terminated processes
-  **Light Blue** — Processes running under the current user
-  **Pink** — Suspended processes



Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
smss.exe	0.12	0.184 K	54,140 K	196		
csrss.exe	74.01	60 K	8 K	0		
System	1.67	68 K	11,380 K	4		
System	1.31	0 K	0 K		via Hardware Interrupts and DPCs	
smss.exe		1,160 K	1,420 K	888		
csrss.exe	< 0.01	2,404 K	10,02,652 K	3056		
smss.exe		2,572 K	8,728 K	1276		
csrss.exe		1,732 K	8,872 K	1384		
services.exe	0.83	7,800 K	20,904 K	1512		
explorer.exe	< 0.01	10,000 K	44,172 K	1672	Host Process for Windows S.	Microsoft Corporation
WinPcap.sys		4,856 K	14,888 K	7204	WinPcap Host	Microsoft Corporation
WinPcap.sys	7.73	23,550 K	31,828 K	7212	WinPcap Host	Microsoft Corporation
smss.exe		2,328 K	11,108 K	7752	Smss to receive asynchronous	Microsoft Corporation
StartMenuExperienceHost.exe	< 0.01	1,44,188 K	1,66,744 K	11404	Windows Start Experience H.	Microsoft Corporation
SearchHost.exe		88,132 K	1,36,876 K	11412		Microsoft Corporation
msedge.exe	< 0.01	43,944 K	1,16,864 K	13064	Microsoft Edge WebView2	Microsoft Corporation
msedge.exe		3,000 K	12,640 K	13188	Microsoft Edge WebView2	Microsoft Corporation
msedge.exe		15,556 K	41,260 K	13456	Microsoft Edge WebView2	Microsoft Corporation
msedge.exe		1,21,770 K	81,208 K	13472	Microsoft Edge WebView2	Microsoft Corporation
msedge.exe		8,804 K	18,388 K	13680	Microsoft Edge WebView2	Microsoft Corporation
msedge.exe		1,37,472 K	1,42,584 K	13696	Microsoft Edge WebView2	Microsoft Corporation
msedge.exe		8,880 K	20,556 K	17036	Microsoft Edge WebView2	Microsoft Corporation
WidgetBoard.exe		1,15,112 K	1,14,388 K	11688		Microsoft Corporation
RuntimeBroker.exe		19,436 K	77,324 K	11624	Runtime Broker	Microsoft Corporation
WidgetService.exe		5,824 K	20,100 K	6976	WidgetService.exe	Microsoft Corporation
smss.exe		1,868 K	8,960 K	11684	Smss to receive asynchronous	Microsoft Corporation
MicrosoftStartFwdHost.exe		10,500 K	42,812 K	15304		Microsoft Corporation
RuntimeBroker.exe		2,436 K	14,168 K	15782	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		3,240 K	22,362 K	15794	Runtime Broker	Microsoft Corporation
TaskHost.exe	11.58	2,32,880 K	2,72,072 K	15700		Microsoft Corporation
RuntimeBroker.exe		2,356 K	13,294 K	12512	Runtime Broker	Microsoft Corporation
IOCC.exe		44,144 K	50,584 K	8396	IOCC	Intel Corporation
csrss.exe		4,560 K	10,504 K	16024	Apple Security Manager	Apple, Inc.
APSDaemon.exe		8,116 K	25,056 K	17004	Apple Push	Apple Inc.
ApplePhotoStreams.exe		8,932 K	41,740 K	21376	iCloud Photo Stream	Apple Inc.
StartExperienceHost.exe	Susp.	82,000 K	87,020 K	22396	Windows Start Experience H.	Microsoft Corporation
RuntimeBroker.exe		8,116 K	42,460 K	20086	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		2,456 K	14,232 K	21186	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		8,532 K	30,472 K	21818	Runtime Broker	Microsoft Corporation

CPU Usage: 26.41% | Commit Charge: 58.66% | Processes: 319 | Physical Usage: 65.24%

Step 3: Identify Suspicious Processes

1. Look for unfamiliar or oddly named processes (e.g., xkdjeo.exe, randomname123.exe).
2. Check the Company Name and Description:
 - Legitimate software usually shows known publishers like *Microsoft*, *Intel*, or *Adobe*.
3. Right-click the process → Properties → go to the Image tab.
4. Verify the Path of the executable file:
 -  **Safe:** C:\Windows\System32\
 -  **Suspicious:** C:\Users\<User>\AppData\Temp\ or Downloads\
5. Check for Digital Signature:
 - Valid signature = trusted developer
 - No signature or invalid = possibly malicious

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
System	0.12	172 K	84,896 K	188		
smss.exe	74.01	8,184 K	54,140 K	228		
System Idle Process	1.87	88 K	11,360 K	4		
System	1.31	0 K	0 K	0		
csrss.exe	< 0.01	2,404 K	10,82,852 K	3096		
services.exe	0.83	7,808 K	20,904 K	1512		
svchost.exe	< 0.01	15,000 K	44,172 K	1672	Host Process for Windows S...	Microsoft Corporation
Wscntfy.exe	4.896 K	14,888 K	7204 K	7204	WMI Provider Host	Microsoft Corporation
Wscntfy.exe	7.73	23,556 K	31,828 K	7212	WMI Provider Host	Microsoft Corporation
smsscap.exe		2,328 K	11,108 K	7752	Sink to receive asynchronous...	Microsoft Corporation
StartMenuExperienceHost.exe	< 0.01	1,44,188 K	1,86,744 K	11404	Windows Start Experience H...	Microsoft Corporation
SearchIndexer.exe		86,132 K	1,36,876 K	11412		Microsoft Corporation
msedge.exe	< 0.01	43,944 K	1,16,864 K	13064	Microsoft Edge WebView2	Microsoft Corporation
msedge.exe		3,000 K	12,840 K	13168	Microsoft Edge WebView2	Microsoft Corporation
msedge.exe		15,556 K	41,260 K	13456	Microsoft Edge WebView2	Microsoft Corporation
msedge.exe		1,21,776 K	81,208 K	13472	Microsoft Edge WebView2	Microsoft Corporation
msedge.exe		9,804 K	16,388 K	13660	Microsoft Edge WebView2	Microsoft Corporation
msedge.exe		1,37,472 K	1,42,564 K	13636	Microsoft Edge WebView2	Microsoft Corporation
msedge.exe		8,868 K	20,556 K	17636	Microsoft Edge WebView2	Microsoft Corporation
WidgetHost.exe	1,15,112 K	1,14,386 K	1,1660			Microsoft Corporation
RuntimeBroker.exe		19,436 K	77,324 K	11848	Runtime Broker	Microsoft Corporation
WidgetHost.exe		5,924 K	29,100 K	8976	WidgetService.exe	Microsoft Corporation
smsscap.exe		1,668 K	9,980 K	11864	Sink to receive asynchronous...	Microsoft Corporation
MicrosoftStartFeedProvi...		10,500 K	42,812 K	15304		Microsoft Corporation
RuntimeBroker.exe		2,436 K	14,168 K	15762	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		3,248 K	22,392 K	15764	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe	0.36	2,32,888 K	2,72,872 K	15760		Microsoft Corporation
RuntimeBroker.exe		2,356 K	13,264 K	12512	Runtime Broker	Microsoft Corporation
igmp.exe		44,144 K	56,684 K	18360	igmp	Intel Corporation
svchost.exe		4,560 K	18,504 K	18524	Apple Security Manager	Apple, Inc.
APSDaemon.exe		6,116 K	25,056 K	17304	Apple Push	Apple Inc.
ApplePhotoStreams.exe		8,932 K	41,740 K	21376	iCloud Photo Stream	Apple Inc.
StartMenuExperienceHost.exe	Stop	62,008 K	87,020 K	22500	Windows Start Experience H...	Microsoft Corporation
RuntimeBroker.exe		8,116 K	42,460 K	20060	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		2,456 K	14,232 K	2100	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		6,532 K	30,472 K	21816	Runtime Broker	Microsoft Corporation

CPU Usage: 26.41% | Commit Charge: 58.66% | Processes: 319 | Physical Usage: 65.24%

Step 4: Analyze Process Behavior

- Observe CPU, Memory, and I/O usage columns.
- If a small or unknown process consumes excessive CPU or memory, it may be malicious.
- Right-click the process → Properties → go to the TCP/IP tab.
- Check if it communicates with unknown external IP addresses.
- Examine Handles and DLLs tabs for suspicious loaded files or libraries.

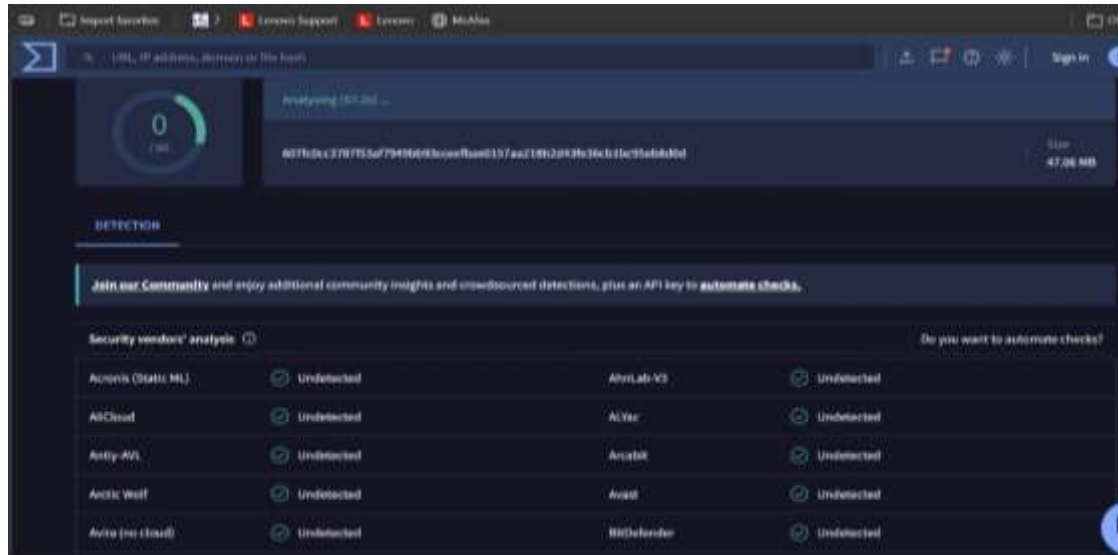
Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
svchost.exe	0.01	148 K	26,63,933 K	148		
svchost.exe		10660		10660		
svchost.exe		12148		12148		
svchost.exe		12572		12572		
svchost.exe		6604		6604		
svchost.exe		11768		11768		
svchost.exe		21456		21456		
svchost.exe		1864		1864		
svchost.exe		6012		6012		
svchost.exe		12988		12988		
svchost.exe		13104		13104		
svchost.exe		2472		2472		
svchost.exe		17052		17052		
svchost.exe		21840		21840		
svchost.exe		10800		10800		
svchost.exe		3204		3204		
svchost.exe		10816		10816		
svchost.exe		2044		2044		
svchost.exe		12324		12324		
svchost.exe		12700		12700		
svchost.exe		12308		12308		
svchost.exe		12312		12312		
svchost.exe		12300		12300		

19:49:20 21-09-2026
Wait:WUserRequest
0:00:07.612
0:00:12.612
95,533
52,60,80,17,360

Step 5: Verify Process Legitimacy

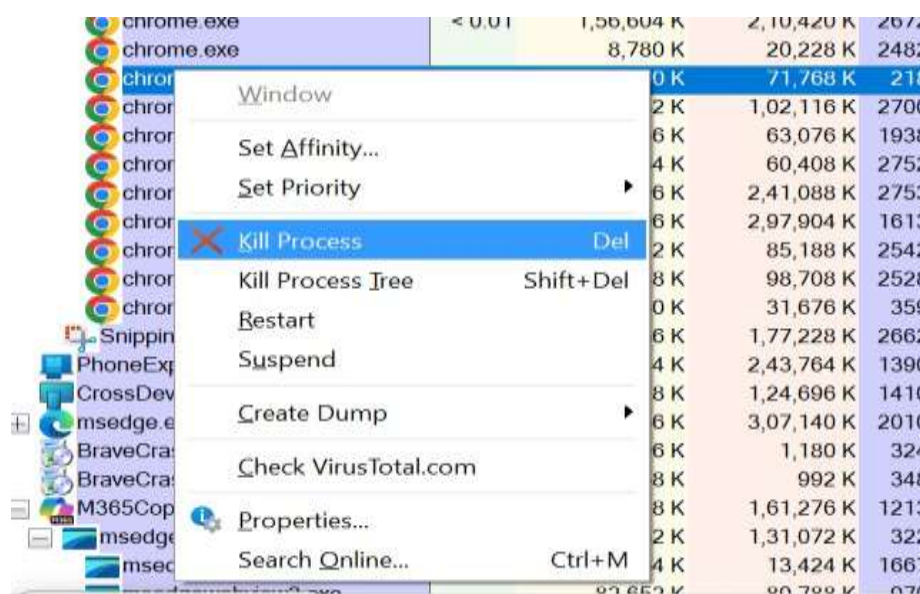
1. Search the process name on Google.
Example: svchost.exe vs svhost.exe (one letter missing — suspicious).

2. Visit  <https://www.virustotal.com>
 - o Upload the process file or search its name to verify if it's reported as malware.
3. Cross-check with **ProcessLibrary.com** or official vendor websites for authenticity.



Step 6: Take Appropriate Action

1. **If the process is confirmed malicious:**
 - Right-click the process → **Kill Process** to stop it.
 - Delete the corresponding executable file from its path.
2. **If unsure:**
 - Right-click → **Suspend Process** to stop it temporarily for investigation.
3. **After removal:**
 - Run a **Full System Scan** using Windows Defender or Malwarebytes to ensure no remnants remain.



Step 7: Example Observation

You find faangpath_simple_template.pdf consuming 70% CPU.

- **Path:** C:\Users\Admin\AppData\Temp\faangpath_simple_template.pdf
- **Digital Signature:** None
- **Company Name:** Unknown
- **Network Activity:** Shows connections to unknown IPs in the TCP/IP tab
- **Online Check:** VirusTotal confirms it as a **known trojan**
- **Action Taken:** Suspended → Killed → Deleted file → Performed full antivirus scan



Result

Using Process Explorer, suspicious processes were successfully identified by examining their **CPU usage**, **path**, **digital signature**, and **network activity**. Confirmed malicious processes were terminated and removed to maintain system integrity.